Python

Recursion

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Agenda

1. what is recursion?
2. Recursion Tree
3. How to approach recursive solution?

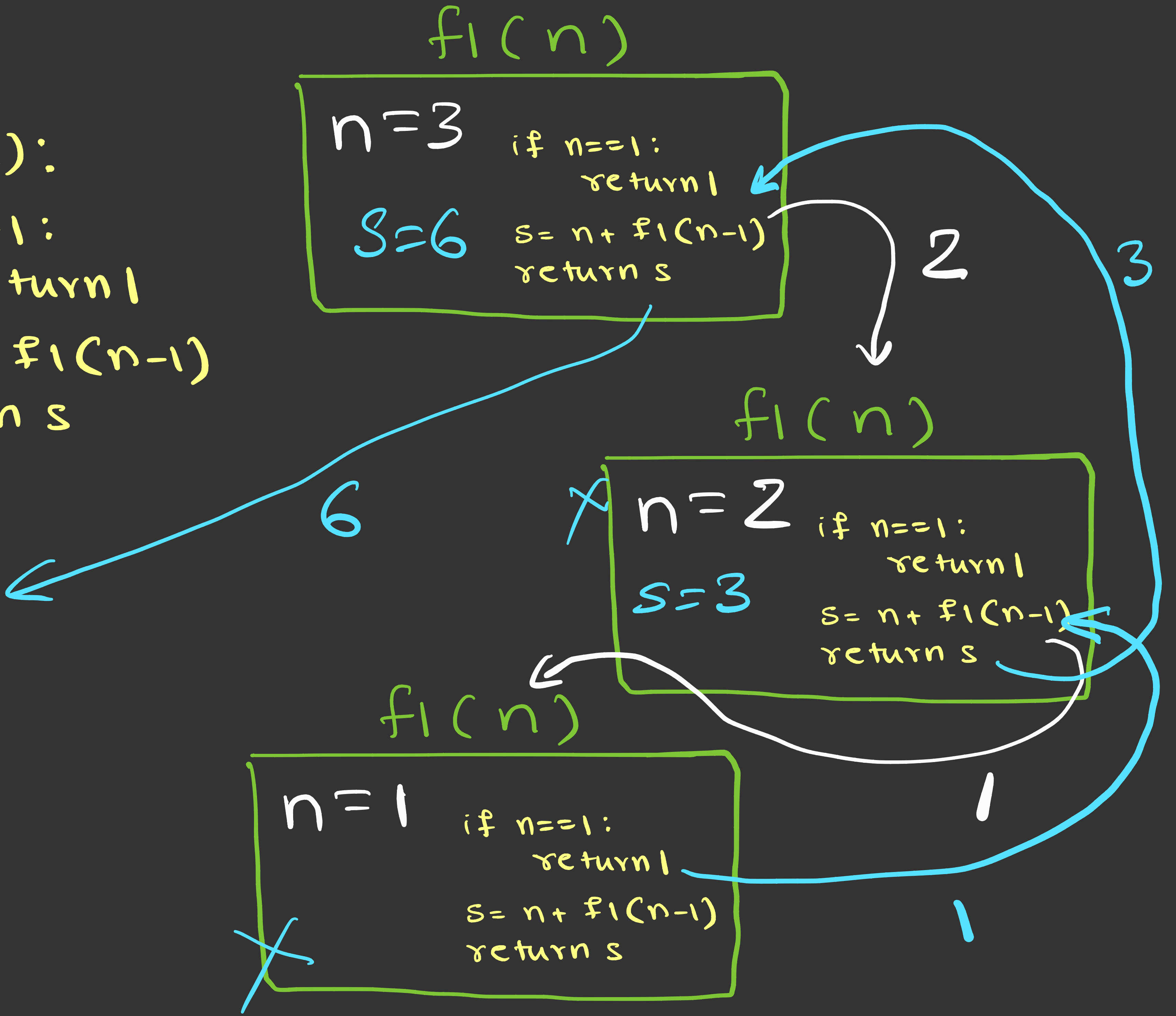
Recursion

Function calling itself is called recursion.

```
def f1():
    ——
    ——
    f1()
    ——
    ——
```

```
def f1(n):
    if n==1:
        return 1
    s= n+ f1(n-1)
    return s
```

```
r= f1(3)
print(r)
```



$f1(3) = 6$

$f(n)$

$f1(10) \rightarrow 10 + f1(9)$

$\hookrightarrow 9 + f1(8)$

$\hookrightarrow 8 + f1(7)$

$7 + f1(6)$

$6 + f1(5)$

$5 + f1(4)$

$\hookrightarrow 4 + f1(3)$

$\hookrightarrow 3 + f1(2)$

$\hookrightarrow 2 + f1(1)$

$\hookrightarrow 1$

$10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$

$$
f1(n) = \begin{cases} 1 & n == 1 \\ n + f1(n-1) & n > 1 \end{cases}
$$

- In recursion, problem is solved in terms of problem itself
- Each time recursive function call to itself for little simpler version of the original problem

$$f1(10) = 10 + f1(9)$$

$$f1(n) = n + f1(n-1)$$

How to approach recursive function?

Write a recursive function to calculate sum of first n natural numbers.

1. sum(n) $1+2+3+4+...+n$

RC 2. n+ sum(n-1) $1+2+3+...+n-1$

BC 3. n==1   1

```
def sum(n):
    if n==1:
        return 1
    return n+ sum(n-1)
```